

Week 3: Seller and Inventory Management System

Scenario

The e-commerce company receives products from different sellers. The system stores the details of each seller and keeps track of the products supplied by them. It also maintains inventory information such as stock quantity and stock status to monitor product availability.

Create Seller and Inventory Tables:

```
SQL> CREATE TABLE Seller (Seller_ID VARCHAR2(7), Seller_Name VARCHAR2(25), Email VARCHAR2(30), Phone NUMBER(10), Address VARCHAR2(45));  
Table created.  
  
SQL> CREATE TABLE Inventory (Inventory_ID VARCHAR2(7), Product_ID VARCHAR2(7), Seller_ID VARCHAR2(7), Stock_Quantity NUMBER(11), Stock_Status VARCHAR2(11), Last_Updated DATE);  
Table created.
```

Establish Relationships

Seller and Product

A seller can supply several products, and a product can be supplied by different sellers. The relationship is maintained through the Inventory table using Seller_ID and Product_ID.

Product and Inventory

Each product has inventory information that records its available stock. The Inventory table stores the stock quantity and stock status for each product.

Primary Key

```
SQL> ALTER TABLE Seller ADD CONSTRAINT PK_Seller PRIMARY KEY (Seller_ID);
Table altered.

SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inventory PRIMARY KEY (Inventory_ID);
Table altered.
```

Foreign Key for Inventory Table:

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT FK_Seller FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID);
Table altered.
```

Unique used for phone number:

```
SQL> ALTER TABLE Seller ADD CONSTRAINT UQ_Seller_Phone UNIQUE (Phone);
Table altered.
```

Not null used for email:

```
SQL> ALTER TABLE Seller MODIFY Email VARCHAR2(30) NOT NULL;
Table altered.
```

Check:

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT CHK_Stock CHECK (Stock_Quantity >= 0);
Table altered.
```

Insert values:

Seller Table:

```
SQL> INSERT INTO Seller VALUES ('S001', 'Arun Traders', 'arun@gmail.com', 9876543210, 'Chennai');
1 row created.

SQL> INSERT INTO Seller VALUES ('S002', 'Kumar Stores', 'kumar@gmail.com', 9876543211, 'Tambaram');
1 row created.

SQL> INSERT INTO Seller VALUES ('S003', 'Priya Enterprises', 'priya@gmail.com', 9876543212, 'Madurai');
1 row created.

SQL> INSERT INTO Seller VALUES ('S004', 'Ravi Mart', 'ravi@gmail.com', 9876543213, 'Coimbatore');
1 row created.
```

```
SQL> SELECT * FROM Seller;
```

SELLER_	SELLER_NAME	EMAIL	PHONE
ADDRESS			
S001	Arun Traders Chennai	arun@gmail.com	9876543210
S002	Kumar Stores Tambaram	kumar@gmail.com	9876543211
S003	Priya Enterprises Madurai	priya@gmail.com	9876543212

SELLER_	SELLER_NAME	EMAIL	PHONE
ADDRESS			
S004	Ravi Mart Coimbatore	ravi@gmail.com	9876543213

Inventory Table:

```
SQL> INSERT INTO Inventory VALUES ('I002', 'P1002', 'S002', 15, 'Available', SYSDATE);  
1 row created.
```

```
SQL> INSERT INTO Inventory VALUES ('I003', 'P1003', 'S003', 10, 'Available', SYSDATE);  
1 row created.
```

```
SQL> INSERT INTO Inventory VALUES ('I004', 'P1004', 'S004', 0, 'Out Stock', SYSDATE);  
1 row created.
```

```
SQL> SELECT * FROM Inventory;
```

INVENTO	PRODUCT	SELLER_	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
I001	P1001	S001	20	Available	20-SEP-26
I002	P1002	S002	15	Available	20-SEP-26
I003	P1003	S003	10	Available	20-SEP-26
I004	P1004	S004	0	Out Stock	20-SEP-26

1.Seller-wise Inventory Report

Display:

- Seller name
- Number of products supplied by each seller
- Total available stock

```
SQL> SELECT s.Seller_Name, COUNT(i.Product_ID) AS Number_of_Products, SUM(i.Stock_Quantity) AS Available_Stock FROM Seller s JOIN Inventory i ON s.Seller_ID = i.Seller_ID WHERE i.Stock_Status = 'Available' GROUP BY s.Seller_Name;
```

SELLER_NAME	NUMBER_OF_PRODUCTS	AVAILABLE_STOCK
Arun Traders	1	20
Kumar Stores	1	15
Priya Enterprises	1	10

2.Stock Availability Report

Display: Product name, stock quantity and stock status.

```
SQL> SELECT p.Product_Name, i.Stock_Quantity, i.Stock_Status FROM Product p JOIN Inventory i ON p.Product_ID = i.Product_ID;
```

PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
Laptops	20	Available
Mobile	15	Available
TShirt	10	Available
Shoes	0	Out Stock

3.Inventory Analysis

Total Available Products:

```
SQL> SELECT COUNT(*) AS Total_Available_Products FROM Inventory WHERE Stock_Status = 'Available';
```

TOTAL_AVAILABLE_PRODUCTS
3

Products Out of Stock:

```
SQL> SELECT COUNT(*) AS OutOfStock_Products FROM Inventory WHERE Stock_Status = 'Out of Stock';
```

OUTOFSTOCK_PRODUCTS
0

Highest Stocked Product:

```
SQL> SELECT p.Product_Name, i.Stock_Quantity FROM Product p JOIN Inventory i ON p.Product_ID = i.Product_ID WHERE i.Stock_Quantity = (SELECT MAX(Stock_Quantity) FROM Inventory);
```

PRODUCT_NAME	STOCK_QUANTITY
Laptops	20

Average Inventory Quantity:

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory FROM Inventory;
```

AVERAGE_INVENTORY
11.25